

Research Article

Engagement in radiation therapy research: What happens after graduation?

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ABSTRACT

Background: To prepare undergraduate radiation therapy (RT) students for the professional role of Scholarly Practitioner the University of Alberta's Radiation Therapy Program (RADTH) provides research education, and students conduct novel research studies during their final practicum year with the final result being a publishable paper. A curriculum evaluation project was carried out to examine the impact of the RADTH undergraduate research education by examining the final outcomes of these research projects and whether the learners carried out further research after graduation.

Methods: Alumni who graduated from 2017 to 2020 were surveyed to seek information on the dissemination of their research projects, whether the projects resulted in a change to practice, policy, or patient care, if subsequent research has been performed by the graduates, and the motivators or barriers to conducting research post-graduation. A subsequent manual search of publication databases was conducted to fill in data gaps pertaining to publications.

Results: All RADTH research projects have been disseminated by conference presentation and/or publication. One project was reported to have had an impact on practice, with no impact reported for five projects and two respondents not sure about any impact. All respondents reported they have not participated in any new research projects since graduation. Barriers listed included: limited local opportunity, lack of topic ideas, other professional development, no interest in research, COVID impact, and lack of research knowledge.

Conclusions: RADTH's research education curriculum successfully enables the RT students to conduct and disseminate research. All

RADTH projects have been successfully disseminated by the graduates. However, participation in research post-graduation is not occurring due to a variety of factors. While MRT education programs are required to develop research skills, this education alone may not alter motivation nor ensure research participation post-graduation. Exploring other avenues of professional scholarship may be key to ensuring contribution to evidence-informed practice.

RÉSUMÉ

Contexte : Pour préparer les étudiants de premier cycle en radiothérapie (RT) au rôle professionnel de praticien érudit, le programme Radiothérapie de l'Université de l'Alberta (RADTH) propose une formation à la recherche, et les étudiants mènent de nouvelles études de recherche au cours de leur dernière année de stage, le résultat final étant un article publiable. Un projet d'évaluation du programme d'études a été mené pour examiner l'incidence de la formation à la recherche du programme RADTH en déterminant si les anciens étudiants s'engagent dans la recherche et sa diffusion après l'obtention de leur diplôme.

Méthodologie : Une enquête a été menée auprès des anciens étudiants ayant obtenu leur diplôme entre 2017 et 2020 afin de recueillir des informations sur la diffusion et l'incidence des projets de recherche du programme, si des recherches ultérieures ont été effectuées par les diplômés et les facteurs de motivation ou les obstacles à la réalisation de recherches après l'obtention du diplôme. Une recherche manuelle des bases de données de publications a ensuite été effectuée pour combler les lacunes des données relatives aux publications.

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Ethical approval: Ethical approval to conduct this study was obtained from the University of Alberta Research Ethics Board 2. Amanda Bolderston is Editor-in-Chief of JMIRS but was blinded to the decision making process.

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Résultats : Tous les projets de recherche du programme RADTH ont été diffusés par le biais de présentations à des conférences et/ou de publications. Un projet a eu une incidence sur la pratique, cinq projets n'ont pas eu d'incidence et deux répondants n'étaient pas sûrs de l'incidence. Toutes les personnes interrogées ont déclaré ne pas avoir participé à de nouveaux projets de recherche depuis l'obtention de leur diplôme. Les obstacles cités étaient les suivants: occasions locales limitées, manque d'idées de sujets, autre développement professionnel, pas d'intérêt pour la recherche, incidence de la COVID et manque de connaissances en matière de recherche.

Keywords: Research education; Radiation therapy education; Scholarly practitioner; Research capacity

Conclusions : Le programme d'enseignement de la recherche du programme RADTH permet aux étudiants en TR de mener et de diffuser des travaux de recherche. Tous les projets du programme RADTH ont été diffusés avec succès par les diplômés. Cependant, la participation à la recherche après l'obtention du diplôme n'a pas lieu en raison de divers facteurs. Bien que les programmes de formation des TRM soient tenus de développer des compétences en matière de recherche, la capacité à traduire ces compétences dans la pratique de la RT des diplômés reste limitée et échappe au contrôle des programmes de formation.

It is well known that research is essential to establish and maintain a profession's evidence base, and that incorporating research evidence into practice is key to improving care and patient outcomes [1,2]. Additionally, a research-active organization can improve performance, including staff satisfaction and retention, increase opportunities for personal and professional growth, and increase the uptake of research evidence into practice [3,4]. The Canadian Association of Medical Radiation Technologists (CAMRT) competency profile describes the practice requirements of medical radiation technologists (MRTs) including the role of Scholarly Practitioner where MRTs "use, share, and where possible, contribute to the MRT body of knowledge through research activities and scholarly practice" (p.5) [5]. In defining scholarship, many nursing and allied health education institutions have adopted the foundational work of Ernest Boyer [6,7]. Boyer's model includes four distinct but interrelated domains of scholarship: Discovery, Integration, Application, and Teaching [8,9]. The scholarship of discovery is often what is emphasized in many healthcare sectors, and is the research focused on theory development and knowledge generation. However, as suggested by Boyer, scholarship encompasses more than traditional research. Thus, many avenues for scholarship exist for MRTs to fulfil their Scholarly Practitioner role.

The dissemination of research plays a critical role in enabling the other scholarly activities of integration, application and teaching. This research forms a body of evidence for others seeking to fill profession-specific practice, policy, and education gaps [10,11]. Thus those conducting research have an obligation to share their findings with the wider community to enable this translation into clinical practice or subsequent research. Researchers working with human subjects have an ethical duty to make their results publicly available, for example by publication and other methods of dissemination [12]. Dissemination is also important for undergraduate MRT students who may "conduct new, innovative research that is imminently applicable in an increasingly evidence-based clinical practice" (p.S11) [13].

As participation in research is an expectation of MRT practice in Canada, all undergraduate students in accredited programs receive education in research methodology to equip them

with the skills to carry out and incorporate research in their clinical practice. Within RADTH's undergraduate curriculum, the theory of research and dissemination is taught primarily using a 3-credit research methodology course (RADTH 401). A detailed research project proposal is developed by students by the end of the course. This research is subsequently carried out in student teams over the final two clinical semesters of the program. The final assessment is a paper written in publishable form. During the program students are encouraged to present their findings at radiation therapy conferences, in particular the RTi3 conference which subsequently publishes all conference abstracts. Publication of the students' final manuscripts after graduation is also strongly encouraged. These forms of dissemination, however, are not currently a course requirement.

The first graduating cohort of students from RADTH's undergraduate program was in 2017. Between 2017 and 2020 12 research projects have been successfully completed by 31 students. A curriculum evaluation project was carried out to examine the impact of the RADTH undergraduate research education. By examining the outcomes of the RADTH research projects and whether the learners carried out further research after graduation, our aim was to answer our research question: Are University of Alberta RADTH alumni engaging in research and its dissemination after graduation?

Methods

An electronic survey was developed and managed using REDCap [14]. The survey link was emailed to all graduates of RADTH between 2017 and 2020. Excluded were those known by the researchers to have left the radiation therapy profession.

An initial survey was piloted by three radiation therapy colleagues to ensure appropriateness and clarity of questions, and logic of electronic survey question branching. The finalized survey consisted of eleven main questions, starting with questions linking each respondent to their research project and graduation year. Subsequent questions sought information regarding the dissemination of findings, the impact on practice, policy or patient care from the completed research projects, and whether any subsequent research has been performed by the graduate. Branching questions probed for specific details about project

dissemination and information about motivators or barriers to conducting research post-graduation. Consent to participate was implied by completion of the survey. The survey remained open for four weeks. Two reminder emails were sent out during this timeframe.

Subsequently a manual search of publication databases was conducted in order to fill in the survey data gaps pertaining to the publication of manuscripts and/or abstracts. Project titles and author names were searched in CINAHL, PubMed, and Google Scholar.

Descriptive statistics were generated from the quantitative survey data. The information retrieved from the publication databases was then cross-referenced with the survey data to provide a more complete picture of the RADTH projects' avenues of dissemination. Thematic analysis of the survey's qualitative data was conducted by both authors.

Ethics approval was obtained from the Research Ethics Board 2 at the University of Alberta (August 2021).

Results

Thirty-one students graduated from RADTH between 2017 and 2020. Two graduates have subsequently left the profession, thus the survey was sent to 29 individuals. Eight surveys were completed resulting in a 28% response rate. These eight respondents represented seven of the 12 completed research projects.

Research project outcomes

Publication of research

Of the seven projects reported by survey respondents, four manuscripts have been published. All were published in the Journal of Medical Imaging and Radiation Sciences (JMIRS). None of the respondents reported having their project abstracts published.

The search of publications found that an additional three RADTH projects have had manuscript publications, resulting in a total 58% publication rate (Fig. 1). Of these seven publications six were in JMIRS and one in Cureus. The publication search also found that seven projects (58%) had abstract publications, four of which survey respondents had indicated no abstract publication. Two projects had both a manuscript and an abstract published. Combined data shows that all RADTH research projects have been published in some way. Fig. 2 outlines publication outcomes by project completion year.

Presentation of research

Results from the survey found that four projects were presented (conference poster or oral presentation) to the radiation therapy community via a provincial conference (1), national conference (3), international conference (1). One project was presented at two conferences. No projects were reported as having been presented at a local inservice or as a webinar. The database search yielded an additional 4 projects presented at the RTi3, a national conference, as evidenced by an abstract

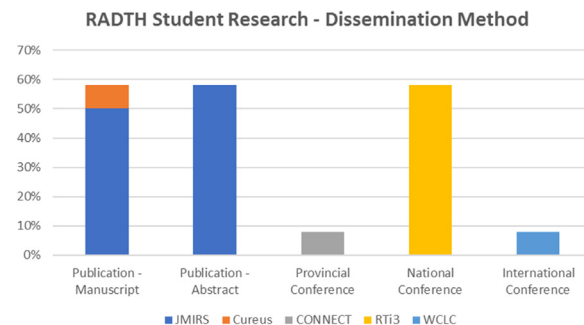


Fig. 1. Dissemination venue and location for the RADTH student research projects. Journal of Medical Imaging and Radiation Sciences (JMIRS), Alberta College of Medical Diagnostic and Therapeutic Technologists CONNECT Conference (CONNECT), University of Toronto, Radiation Oncology RTi3 Conference (RTi3), International Association for the Study of Lung Cancer World Conference on Lung Cancer (WCLC).

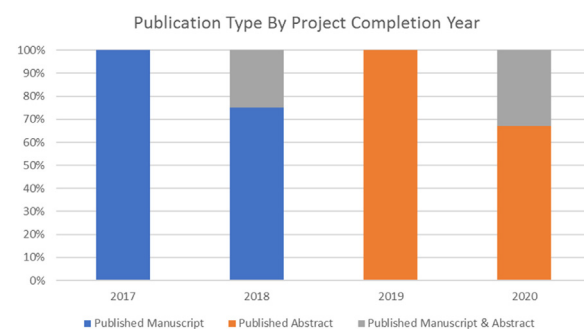


Fig. 2. Publication type by RADTH student research project year.

publication. Fig. 1 outlines the conferences at which these were presented.

Impact on practice

When asked whether their research resulted in any known change in practice, policy or patient care, no change was reported for four projects by five respondents. One project was reported to have resulted in a practice change by one respondent. Two respondents were unsure whether their research project had an effect.

Continuation of research activities post-graduation

Seven respondents indicated that no further research on their project topics has occurred by any of the student team members. One respondent was unsure if the project topic continued to be investigated by any other team member.

All respondents reported they have not participated in any new research projects since graduation.

Responses to the open question "What are some reasons why you have not pursued more research?" elicited comments which grouped into six themes.

1. Limited local opportunities

Several respondents mentioned that their centre, or their role, limited their access to research opportunities:

"Casual and temp positions did not grant me access to research opportunities as I was not permanent staff."

One respondent also commented that they were unsure how to access or get involved in research at their centre.

"I currently work at a smaller centre that does not have many resources nor staff capacity for further research projects."

2. Lack of research topic ideas

Two respondents felt that they had not encountered anything so far in their practice that had sparked research passion:

"I lack ideas. Haven't heard of any projects that especially interest me or I feel passionate about."

3. Pursuing professional development opportunities

A number of respondents told us that they had undertaken additional professional development, such as post-graduate degrees or certifications. This education had left little time for research activities.

"Other opportunities arose e.g. I'm currently taking (course at local college) on top of full-time work."

4. No interest in research

One prominent theme was a lack of interest in research. One respondent additionally commented that their "personal life" was more of a priority. Two people mentioned that they were happy with just the "patient interaction" element of the role.

"Personal life is a priority over research - Very content simply just being an RT."

"I get lots of enjoyment out of the patient interaction side of things and don't want this to be taken away/this viewpoint to change."

5. COVID-19 pandemic

The effects of the COVID-19 pandemic have affected research activity.

"Work has been exhausting especially during the pandemic."

6. Lack of research knowledge

Finally one respondent indicated that they needed further education in how to conduct research.

"Truthfully, I didn't feel like I properly really learnt how to do research, and don't want to put the time into learning how to do it now."

Discussion

RADTH aims to equip learners with the skills and knowledge to be scholarly practitioners in their future careers; to develop the confidence and ability to search published literature sources, conduct clinical research projects, and communicate findings to their peers and the local RT community. To evaluate the impact of the research education the RADTH students receive, this study sought to determine whether the RADTH graduates are engaging in research and research dissemination

by examining if or how their undergraduate research projects were disseminated, and whether they continued to participate in research activities in their professional careers.

Fifty-eight percent of the student projects in this study were published in a peer-reviewed journal after graduation. This is a relatively high number when compared to publication rates for other healthcare student research. Studies examining publication of medical students' research find rates that vary from 8 to 27.7% [15,16,17]. In allied health, 18% of physiotherapy graduate students and 42.3% of pharmacy doctoral students were found to have published their research projects in peer-reviewed journals [18,19]. However, the extent to which other allied health undergraduate research results, including radiation therapy, are disseminated scientifically remains unknown.

Conference submissions are often published as abstracts that are searchable in electronic databases. This is usually seen as a step towards full (journal) publication, because the peer-review process for conference abstracts is often limited and the abstracts are short and lack detail. However, more than half of the studies presented in conference abstracts fail to be published in full [20]. In Canada, the largest annual conference with a radiation therapy focus is RTi3 [21]. RTi3 encourages student submission with a specialized stream that allows students to submit "works in progress" and all abstracts accepted at RTi3 are subsequently published in the JMIRS. All abstract publications of the RADTH research projects in this study represented presentations at RTi3 as either an oral presentation or a poster. This type of publication may not have been evident to survey respondents, as no one indicated they had abstract publications despite indicating they had presented at RTi3. In a short communication McGuffin found that approximately 39% of student abstracts from the 2013 RTi3 conference were subsequently published [13]. It is a lengthy process to convert a conference presentation to a published paper therefore publication numbers may increase beyond the time span of this project.

Respondents were asked whether their research resulted in any known change in practice, policy or patient care in their radiation therapy departments. Most said that their work had not; although this may be hard to determine if respondents are no longer working in the same department that they conducted the research, as was the case with many of this study's participants.

All respondents indicated that they have not pursued any further research, either related to their RADTH project or in any new areas. There is little literature examining the topic of alumni involvement in research after graduation in the medical radiation sciences. However, one study of graduates of a radiologic technology degree program in Uganda found that 76% of their respondents (graduates) have been involved in research in some form after graduation and 50% would be interested in doing future research [22]. Post-graduation research activities listed included "operations research to improve routine work practice", "radiography-related clinical research", "multidisciplinary research with other professions" and "writing projects" (p.116) [22]. This is quite a different result from our study and may be due to a number of factors.

The transition from student to qualified MRT

Transitioning from a student role to that of a newly qualified MRT can be a stressful and challenging time [23,24]. Turner et al. have speculated that new MRTs may be preoccupied with their new professional responsibilities, including keeping up with evolving technology, which may leave little mental capacity for involvement in research activities [25]. Indeed, a number of our survey respondents reported that additional professional development post-graduation precluded their involvement in research. All RADTH graduates in this study had been working for four years or fewer thus may still be in the transitioning period of their careers.

Mentorship

RADTH students are matched with a mentor within their clinical practicum departments. Mentorship is key in supporting novice researchers and can help generate research ideas, provide advice and encouragement and assist in final publication [26]. A lack of topic ideas and research knowledge were two themes that emerged as research barriers in this study; both of which could be mitigated by having readily available mentors within the radiation therapy department. In a relatively young academic profession such as radiation therapy, mentorship may have to come from the other professionals working in radiation oncology such as medical physicists or radiation oncologists [27].

Research interest

A prominent theme in the qualitative data was simply a lack of interest in research. In a CAMRT engagement survey, only 32% of MRT respondents were interested in research [28]. However, a Canadian survey by Moran and Davis found 69% of radiation therapists had participated in research, potentially indicating a higher interest and/or involvement in research for radiation therapists [29]. Unfortunately, this paper does not make it clear whether this participation was at undergraduate level or after graduation. Moran and Davis additionally comment that intrinsic factors like passion and enquiry are perhaps the most likely indicator of research involvement after graduation. Researchers may have to dedicate personal time to research outside work hours, as mentioned by one of the study respondents. Research in many centres may only be undertaken by “a few initial innovators and early adopters” unless departmental culture includes enabling elements (p.171) [30]. We can speculate that a lack of interest may be exacerbated by a departmental culture that does not value research.

Research readiness

One respondent commented that they felt they had not received enough education about research. Moran and Davis found that only 5% of respondents strongly agreed that they received sufficient undergraduate education to develop a successful research project [29]. This is reflected in a number of

other studies that conclude a lack of knowledge is a barrier to RT research activity [25,31,32].

Department culture

Department research culture is also a likely contributing factor. Research engagement will be dependant on a workplace culture which includes being valued by leadership and supported by peers, the availability of mentorship, and having a collaborative work environment [29,33].

Limited local opportunities were mentioned by several respondents who stated that their centre, or their role, limited their access to research opportunities. For research to be a priority, organizations and workplaces must ensure that research is considered core business across individual, organizational and policy levels [11,34]. It also demands that equal value is placed by departments on technical skills that enable new graduates to “hit the floor running” and skills that allow them to “hit the floor thinking” and enable engagement in research after graduation (p.80) [35]. However, while the CAMRT competency profile states that all Canadian radiation therapy graduates must be research proficient when entering the workforce [5], Rosewall et al. have commented that this undergraduate research education has limited impact on subsequent research output without additional workplace research strategic planning [30].

Effects of the COVID-19 pandemic

The COVID-19 pandemic has significantly impacted radiation therapy global work practices, patient care and research activities [36]. Additionally, higher levels of radiation therapist stress and departmental staffing changes (such as deployment and shortages) have been common [37]. It is therefore likely, as one respondent commented, that the pandemic has also affected radiation therapists’ time and energy to carry out research.

Implications of findings to RADTH

The findings of this study show that RADTH’s research education curriculum successfully enables the RT students to conduct and disseminate research. While over half of the student projects were published in a peer-reviewed journal after graduation as a full manuscript (as opposed to a conference abstract), there is still room for improvement. Planned support to increase full manuscript publication rates includes the introduction of revised mentorship agreements that state the expectation of post graduation publication, adding more resources for the process of journal submission, and encouraging graduates to seek mentorship from the JMIRS editorial staff.

While the results of this study demonstrated that none of the participants were carrying out novel research after graduation, the CAMRT role of Scholarly Practitioner involves a wider range of skills. It is hoped that our undergraduate research education forms the foundation of all scholarly activities undertaken by the graduates in their professional lives. Going

forward, the research education curriculum will focus on the broader applications of scholarship within RT practice, including quality improvement activities, critical appraisal of literature, and the integration of research into practice.

Future research

Future research directions could include investigating whether RADTH graduates are engaging in other areas of scholarly activity in radiation therapy practice, as this was not addressed in this study. A national perspective of the impact of undergraduate research education could be obtained by conducting a similar study with other Canadian RT education programs. This could allow sharing of best practices and inform broader changes in curricula.

Limitations

This study was limited by the low survey response rate. Additionally, while missing data regarding publications of the student research was retrieved by the manual literature search, information on other research dissemination avenues for five RADTH projects was missing.

This study sought information from RADTH graduates who have been in the workplace for less than five years; with the 2020 graduation cohort having graduated only one and half years prior to the study survey. It is possible that the value of the research education on later career research has yet to be realized.

The term “research” was implied to refer to novel research studies only, with survey questions asking about contributions to new research projects and their dissemination. Other scholarly activities related to, for example, evidence-based practice and quality improvement projects were not specifically assessed. It is possible that RADTH graduates are demonstrating their professional role of Scholarly Practitioner in alternative ways and the research education they received may have motivated and contributed to their ability to do so.

Conclusion

While all student projects in this study were disseminated in some way, it was also found that no students were engaging in research after graduation. Additionally, many of the same barriers cited in other studies were found, such as limited local opportunities, topic ideas or lack of interest. Thus the provision of undergraduate research education alone does not appear to alter motivation nor ensure participation in research activities post-graduation for this particular population. This study adds the novel findings that, despite successfully disseminating undergraduate research, barriers to further engagement may reflect those seen in the broader literature.

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References

- [1] Higgins J, Davey C, Li W, Chan K, Wenz J, Rosewall T. 10 years of exposure to a radiation therapist research culture: where are we now? *J Med Imaging Radiat Sci*. 2011;42(3):106–112. doi:10.1016/j.jmir.2011.04.004.
- [2] Harding K, Lynch L, Porter J, Taylor NF. Organisational benefits of a strong research culture in a health service: a systematic review. *Austral Health Rev*. 2017;41(1):45–53. doi:10.1071/AH15180.
- [3] Dennett AM, Cauchi T, Harding KE, Kelly P, Ashby G, Taylor NF. Research interest, experience and confidence of allied health professionals working in medical imaging: a cross-sectional survey. *J Med Radiat Sci*. 2021;68(2):121–130. doi:10.1002/jmrs.432.
- [4] Strudwick R, Harris M, McAlinney H, et al. The College of Radiographers Research Strategy for the next five years. *Radiography*. 2021;27(Suppl 1):S5–S8. doi:10.1016/j.radi.2021.06.010.
- [5] . *CAMRT National Competency Profile for Entry-Level MRTs in Canada*; 2019. <https://www.camrt.ca/wp-content/uploads/2021/10/National-Competency-Profile-2019.pdf>.
- [6] Coulton Kimberly Mathieu *Academic Tenure: defining Scholarship in the Health Professions*. 392; 2011 Electronic Theses and Dissertations <https://digitalcommons.georgiasouthern.edu/etd/392>.
- [7] Berry LE. Furthering nursing scholarship in today's postsecondary institutions: the legacy of Ernest Boyer. *Qual Adv Nurs Educ*. 2015;1(3). doi:10.17483/2368-6669.1038.
- [8] Boyer EL. *Scholarship reconsidered: priorities of the professoriate*. Carnegie Foundation for the Advancement of Teaching; 1990.
- [9] Cleland JA, Jamieson S, Kusurkar RA, Ramani S, Wilkinson TJ, van Schalkwyk S. Redefining scholarship for health professions education: AMEE Guide No. 142. *Med Teach*. 2021;43(7):824–838. doi:10.1080/0142159X.2021.1900555.
- [10] Bolderston A. Five percent is not enough! Why we need more qualitative research in the medical radiation sciences. *J Med Imaging Radiat Sci*. 2014;45(3):201–203. doi:10.1016/j.jmir.2014.06.006.
- [11] Matus J, Walker A, Mickan S. Research capacity building frameworks for allied health professionals - a systematic review. *BMC Health Serv Res*. 2018;18(1):716. doi:10.1186/s12913-018-3518-7.
- [12] World Medical Association. *WMA Declaration of Helsinki - Ethical Principles for Medical Research Involving Human Subjects*. Accessed 17 March 2022. Available at: <https://www.wma.net/policies-post/wma-declaration-of-helsinki-ethical-principles-for-medical-research-involving-human-subjects/>
- [13] McGuffin M. Growing the professional body of knowledge through undergraduate student research: can we get that in writing? *J Med Imaging Radiat Sci*. 2015;46(3):S10–S12. doi:10.1016/j.jmir.2015.06.008.
- [14] Harris PA, Taylor R, Thielke R, Payne J, Gonzalez N, Conde JG. Research electronic data capture (REDCap) - a metadata-driven methodology and workflow process for providing translational research informatics support. *J Biomed Inform*. 2009;42(2):377–381. doi:10.1016/j.jbi.2008.08.010.
- [15] Rhyne RL. A scholarly research requirement for medical students. The ultimate problem based learning experience. *Acad Med*. 2000;75(5):523–524.
- [16] Arriola-Quiroz I, Curioso WH, Cruz-Encarnacion M, Gayoso O. Characteristics and publication patterns of theses from a Peruvian medical school. *Health Info Libr J*. 2010;27:148–154. doi:10.1111/j.1471-1842.2010.00878.x.
- [17] den Bakker CR, Ommering BWC, van Leeuwen TN, Dekker FW, Jan de Beaufort A. Assessing publication rates from medical students' mandatory research projects in the Netherlands: a follow-up study of 10 cohorts of medical students. *BMJ Open*. 2022;12:e056053. doi:10.1136/bmjopen-2021-056053.
- [18] Nunes GS, Adami SL, Pellenz MM, et al. Toward scientific dissemination of undergraduate thesis in physical therapy programs - a cross-sectional study. *BMC Med Educ*. 2022;22(1):32. doi:10.1186/s12909-021-03087-8.

- [19] Assemi M, Ibarra F, Mallios R, Corelli RL. Scholarly contributions of required senior research projects in a doctor of pharmacy curriculum. *Am J Pharm Educ*. 2015;79(2):23. doi:10.5688/ajpe79223.
- [20] Scherer RW, Meerpohl JJ, Pfeifer N, Schmucker C, Schwarzer G, von Elm E. Full publication of results initially presented in abstracts. *Cochrane Database System Rev*. 2018(11):MR000005. doi:10.1002/14651858.MR000005.pub4.
- [21] RTi3 Conference (n.d). Radiation Oncology, University of Toronto. <https://radonc.utoronto.ca/rti3-conference>
- [22] Mubuuke AG, Businge F. Self-reported competence and impact of research training among medical radiography graduates from a developing country. *J Med Imaging Radiat Sci*. 2019;50(1):113–118. doi:10.1016/j.jmir.2018.09.002.
- [23] Harvey-Lloyd JM, Morris J, Stew G. Being a newly qualified diagnostic radiographer: learning to fly in the face of reality. *Radiography*. 2019;25(3):e63–e67. doi:10.1016/j.radi.2019.01.007.
- [24] Tan K, Feuz C, Bolderston A, Palmer C. A literature review of preceptorship: a model for the medical radiation sciences? *J Med Imaging Radiat Sci*. 2011;42(1):15–20. doi:10.1016/j.jmir.2010.08.004.
- [25] Turner A, D'Alimonte L, Fitch M. Promoting radiation therapy research: understanding perspectives, transforming culture. *J Radiother Pract*. 2013;12:92–99. doi:10.1017/S1460396912000088.
- [26] D'Alimonte L. Why student research sparks passion! Perspectives from a clinical scientist. *J Med Imaging Radiat Sci*. 2016;47((3), Supplement) S8–9. doi:10.1016/j.jmir.2016.07.007.
- [27] Velec M. Our research output is up, but is the quality? *J Med Imaging Radiat Sci*. 2016;47(3):213–216. doi:10.1016/j.jmir.2016.07.006.
- [28] Canadian Association of Medical Radiation Technologists (CAMRT). Research engagement survey results. *CAMRT News*. 2018;36(3):22.
- [29] Moran D, Davis C. Pan-Canadian survey of medical radiation technologists' views toward evidence-based practice, research, barriers, and enablers. *J Med Imaging Radiat Sci*. 2020;51(1):29–39.
- [30] Rosewall T, Kelly V, Higgins J, et al. The influence of programmatic change on radiation therapist research capacity—a single-center case study. *J Med Imaging Radiat Sci*. 2009;40(4):170–177. doi:10.1016/j.jmir.2009.08.002.
- [31] Agustin C, Grand M, Gebiski V, Turner S. Radiation therapists' perspective on barriers to clinical trials research. *J Med Imaging Radiat Oncol*. 2008;52(2):178–182. doi:10.1111/j.1440-1673.2008.01938.x.
- [32] Halkett GKB, Berg M, Ebert MA, et al. Radiation therapists' perspectives on participating in research. *J Med Radiat Sci*. 2017;64(4):299–309. doi:10.1002/jmrs.237.
- [33] Higgins R, Robinson L, Hogg P. Unlocking student research potential: toward a research culture in radiography undergraduate learning curricular. *J Med Imaging Radiat Sci*. 2015;46((3), Supplement) S6–S9. doi:10.1016/j.jmir.2015.06.009.
- [34] Neep MJ. We can do it: improving research culture and capacity in medical radiation sciences. *J Med Radiat Sci*. 2021;68(2):105–107. doi:10.1002/jmrs.470.
- [35] McKnight KL. Research pedagogy in a UK radiography education setting. *Radiography*. 2022;28(1):80–87. doi:10.1016/j.radi.2021.08.002.
- [36] Boon IS, Lim JS, Au Yong TPT, Boon CS. Digital healthcare and shifting equipoise in radiation oncology: the butterfly effect of the COVID-19 pandemic. *J Med Imaging Radiat Sci*. 2021;52(1):11–13. doi:10.1016/j.jmir.2020.10.002.
- [37] Morassaei S, Di Prospero L, Ringdalen E, et al. A survey to explore the psychological impact of the COVID-19 pandemic on radiation therapists in Norway and Canada: a tale of two countries. *J Med Radiat Sci*. 2021;68(4):407–417. doi:10.1002/jmrs.557.